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```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define MAX_ITEMS 5

// Shared buffer
int buffer;

// Semaphores
sem_t empty;
sem_t full;

// Producer Thread
void* producer(void* arg)
{
    for(int i = 1; i <= MAX_ITEMS; i++)
    {
        // Wait until buffer becomes empty
        sem_wait(&empty);

        buffer = i;

        printf("\n[Producer] Produced item %d\n", buffer);

        sleep(1);

        // Signal consumer that item is available
        sem_post(&full);
    }

    pthread_exit(NULL);
}

// Consumer Thread
void* consumer(void* arg)
{
    for(int i = 1; i <= MAX_ITEMS; i++)
    {
```

```

        printf("[Consumer] Waiting for item...\n");

        // Wait until item becomes available
        sem_wait(&full);

        printf("[Consumer] Consumed item %d\n", buffer);

        sleep(1);

        // Signal producer that buffer is empty
        sem_post(&empty);
    }

    pthread_exit(NULL);
}

int main()
{
    pthread_t producerThread;
    pthread_t consumerThread;

    // Initialize semaphores
    sem_init(&empty, 0, 1);
    sem_init(&full, 0, 0);

    printf("=== Producer Consumer Synchronization ===\n");

    // Create threads
    pthread_create(&producerThread, NULL, producer, NULL);
    pthread_create(&consumerThread, NULL, consumer, NULL);

    // Wait for threads to finish
    pthread_join(producerThread, NULL);
    pthread_join(consumerThread, NULL);

    // Destroy semaphores
    sem_destroy(&empty);
    sem_destroy(&full);

    printf("\nAll threads finished execution.\n");

    return 0;
}

```

OUTPUT:

```
PS C:\Users\prerana\OneDrive\Desktop\pthread_program> .\producer_consumer.exe
Producer produced item 2 | Buffer = 1
Consumer consumed item 2 | Buffer = 0
Producer produced item 3 | Buffer = 1
Producer produced item 4 | Buffer = 2
Consumer consumed item 3 | Buffer = 1
Producer produced item 5 | Buffer = 2
Producer produced item 6 | Buffer = 3
Consumer consumed item 4 | Buffer = 2
Producer produced item 7 | Buffer = 3
Producer produced item 8 | Buffer = 4
Consumer consumed item 5 | Buffer = 3
Producer produced item 9 | Buffer = 4
Producer produced item 10 | Buffer = 5
Consumer consumed item 6 | Buffer = 4
Consumer consumed item 7 | Buffer = 3
Consumer consumed item 8 | Buffer = 2
Consumer consumed item 9 | Buffer = 1
Consumer consumed item 10 | Buffer = 0
PS C:\Users\prerana\OneDrive\Desktop\pthread_program> |
```